

SIRATCHA BOONSAKDEE

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EDUCATION

Singapore University of Technology and Design (SUTD)

Sep 2024 – Present

B.Eng. Engineering Product Development, Electrical Engineering Specialization | Minors: Psychology & Business Management, Computer Science

- GPA 4.35/5.00 · Kewalram Chanrai Group Scholarship for Women · SEED-2 Program (Canada)

Global Exchange Programme (GEXP) — University of Waterloo, Faculty of Engineering

Sep – Dec 2026

- Selected for a semester-long exchange, supported by a CAD 10,200 SEED-2 scholarship.

WORK EXPERIENCE

Chulalongkorn University — Research Internship

Apr – May 2023

- Conducted DOE-based testing to resolve sensor performance variances; analyzed SEM/TEM/EDS data; earned Regional Silver Medal, NSTDA Young Scientist Competition.

WorkX — Business Development Internship

Sep – Dec 2025

- Conducted market analysis and customer segmentation for an AI startup; produced strategy documents and a go-to-market framework.

SELECTED ENGINEERING PROJECTS

Retail Shelf Monitoring Robot

May – Aug 2026

- Built the moving base (ROS-based SLAM navigation) with custom PCB and power management; TVS diode for transient voltage protection; oscilloscope/multimeter debugging; maintained BOM.
- Integrated multi-device communication across USB Serial and Wi-Fi/HTTP REST API (JSON) between Raspberry Pi, Arduino, ESP32, LiDAR, and a networked camera unit.

DynaGuard Smart Knee Brace

Mar – May 2026

- ESP32-based closed-loop pressure control system integrating FSR402 force sensor, MPU6050 IMU, and N20 motor actuator (MX1508 driver); 4-person team; achieved real-time strap adjustment.

Solar-Powered In-Car Air Purifier

Jun – Aug 2025

- Designed a battery-based power distribution system integrating solar charging, Li-ion storage, and DC load management; applied IV curve analysis and battery sizing; validated via CAD-modeled enclosure and data-driven testing.

Cyber-Clamp — Smart Locking System

Feb – Apr 2025

- Designed a modular RFID-based smart-locking system for SUTD's Fab Lab, replacing an untracked manual tool-borrowing process with automated check-in/check-out and live status updates; modeled the locking mechanism and casing in CAD and programmed the Arduino to control the RFID reader, IR sensor, and motor-driven lock.

Diffraction Grating Design

May – Aug 2026

- Designed and fabricated a diffractive-grating structure on silicon in SUTD's cleanroom, applying a Design of Experiments (DOE) approach to optimize the photolithography and patterning process.

TECHNICAL SKILLS

PCB & Hardware: KiCad, PCB assembly & soldering, schematic drafting

Embedded Systems & Comms: Arduino, ESP32/ESP32-S3, Raspberry Pi, C, C++, Python, USB Serial, Wi-Fi/HTTP REST API, JSON

Test & Debug: Oscilloscope/multimeter debugging, sensor calibration, test documentation

Power Systems: Battery-based power design, DC load/power budgeting, BOM creation & cost tracking

Tools: Fusion 360, Multisim, DOE, Figma

Languages: Thai (Native), English (Proficient), Chinese (Beginner)

LEADERSHIP

President, SUTD Beauty of Thai Interest Group

May 2025 – Mar 2026

- Led operations and events promoting Thai culture, engaging 50+ participants.

Welfare Director & Publication Head, SUTD Ambassador Club

May 2025 – Mar 2026

- Managed club communications; led branding and presentation design for external outreach; delivered presentations to 100+ participants.

EXTRACURRICULAR ACTIVITIES

IC Design Summer Camp 2026

Jul 2026

- Gained exposure to analog/digital IC design at AMD and Marvell, and explored semiconductor manufacturing processes at STMicroelectronics.

SEMI SEA Tech Zoomer Regional Bootcamp

May 2026

- Served as Discussion Lead on a data-driven packaging reliability risk prediction project under Micron mentorship; industry talks and site visit to MIMOS Berhad.

SUTD Translating Innovative Intelligent Design to Entrepreneurship Programme

Sep – Dec 2025

- Selected for a 3-month exchange at Zhejiang University studying AI tool applications and digital business strategy, supported by a SGD 4,000 SUTD scholarship.